

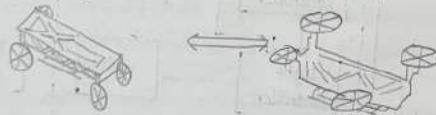
Group code: MNN-10 Student names: Harshil Singla, Arin Waling, Shriti Adhi, Meel Ramthia, Saumya EDL 2025
Project title: Auto-transforming Robot Toy ^{Shan} Date: _____

Use your notebooks for discussions and rough work. Fill out this sheet after working individually and discussing within your team.

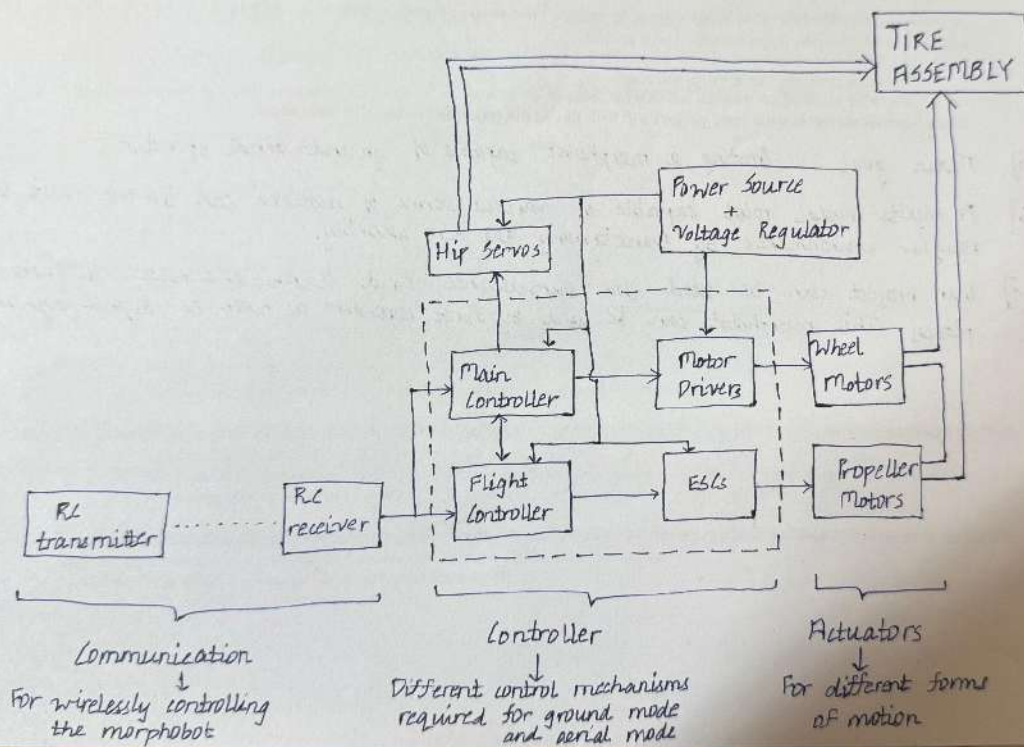
1. In simple words, describe what you are going to build in your project, what its purpose is, and how it will function. Be as detailed as possible, covering all the major aspects of your project.
- What is the main goal of your project?
 - What problem does it solve, and how?
 - Who will use your project, and in what context?

Draw a pencil sketch of what your project will look like at the end of the course, for final demo.

- Main goal → creating a morphobot capable of ground+aerial operations
- A multi-modal robot capable of various forms of motions can traverse vast terrains, with rougher environments, by transforming the tire assembly.
- Our project can be used for surveillance and search-and-rescue in hard-to-reach places. This morphobot can be used by first responders as well as defence agencies.



2. Draw a block diagram of your project. Create a visual representation showing the key components or subsystems of your project. For each block in the diagram, briefly explain its main function and how it fits into the overall system.
- What are the main subsystems or modules of your project?
 - How do they interact with each other?



MDN-10

EDL-11

3. Write down details for these blocks: What are the key performance metrics for each block (e.g., power, size, speed)? What trade-offs are you considering in your design choices? Are there any constraints or limitations for each block?

Block	Key specifications of this block	Design choices for this block
Communication module	RC receiver and transmitter based on manual control (joystick) → Camera feed	- cost-effective - ~200m range
Controller module	- distinct main controller and flight controller - ESC, motor drivers (x4) sensor unit	- land motion and transformation by main controller - compact and compatible with flight controller software
Power module	- Servo motors (x4) - wheel motors (x4) - propellers (x4) - battery (24V)	- weight constraints - for transformation - vehicular / road motion - aerial motion - less weight, high thrust
Mechanical (chassis)	- carbon-fibre	- lightweight, design to assist aerodynamics
(Gear Box)	- gear mesh for Hitec assembly	- robust - torque and centre of mass requirements distribution

4. **What are the unknowns or uncertainties in this project?** Identify aspects of your project that you are uncertain about or that require further research. This may include areas where you know what you need to do but are unsure how to approach it.
- What technical challenges or questions are you facing?
 - Are there any assumptions you must make in order to move forward?

- a) **Mechanical Assembly** - Backlash in motors, Torque Requirements, Mass Distribution
Communications - Robustness / Reliability of video feed
Manufacturing / Material Selection - Lightweight but strong material required, 3-D Printing Limitations
Which components can we make and which to buy?
- b) We can manufacture everything.
We can obtain a basic CAD design to work upon.

Other things to consider from now until Milestone 1 deadline:

5. **Roles and Responsibilities: How will the work be divided among team members?** Assign specific tasks and responsibilities to each team member. Be clear about who is responsible for each part of the project.
- Who will work on which blocks or subsystems?
 - What are the deadlines for each task?
 - How will the team communicate and coordinate to ensure everyone is on track?
6. **Next Steps: What is your plan for the next phase of the project?** Outline what needs to be done in the short-term to move forward.
- What are the immediate next tasks or priorities?
 - Are there any dependencies between tasks? How will you handle these interdependencies?
 - What resources or materials do you need to proceed?
7. **Feedback and Collaboration: How will you gather feedback and collaborate during the project?** Describe how your team plans to share progress, give and receive feedback, and collaborate throughout the course of the project.
- How often will you check in with your team members?
 - Will you conduct regular brainstorming or review sessions?